

Diego Aguilar

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EDUCATION

Bachelor of Science in Computer Science

August 2022 - May 2026

University of Florida, Gainesville, FL

- **GPA:** 3.73 / 4.00
- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Object-Oriented Programming, Operating Systems, Algorithm Abstraction, Programming Language Concepts, Network Fundamentals, Information & Database Systems.

PROFESSIONAL EXPERIENCE

Backend Software Engineer

December 2024 - Present

ServeU, Gainesville, FL

- Developed optimized backend systems for a customer-facing web app, focusing on efficient data processing and performance.
- Collaborated with a team and led discussions to design and implement a functional web app to showcase to investors.
- Led the migration from Flask to Firebase, reducing hosting costs by 90% and simplifying the frontend to backend communication.

Teaching Assistant

September 2024 - Present

Dr. Laura Cruz Castro | University of Florida, Gainesville, FL

- Worked directly with the professor of a python course and a graduate data science course to develop and test curriculum materials.
- Analyzed course content to develop lecture slides, homework assignments, and assist in the creation and revisions of exams.

Teaching Assistant

January 2023 - Present

COP3503C | University of Florida, Gainesville, FL

- Led weekly discussions to 30+ students, teaching core computer science topics and demonstrating their practical applications.
- Developed an auto-grader in Gradescope for students' capstone project, reducing semester workload by 100+ hours.
- Directed Peer Mentor teams to enhance assignment design and develop interactive lab materials, fostering improved student engagement and understanding of computer science concepts.

PROJECTS

Minesweeper Autograder

September 2024

COP3503C - TA | University of Florida, Gainesville, FL

- Piloted an auto-grading library that when linked to student submissions, produces a score and point breakdown.
- Leveraged design patterns to keep track of code state and strategically organized the large code base of the library.

Disaster Family

June 2024

CEN4930 | University of Florida, Kyoto, Japan

- Collaborated with a team to create a game focused on Cross Cultural Design with the theme of disaster prevention.
- Integrated core mechanics utilizing design patterns, including decorators, commands, and mediators.

Advisor Ally

January 2024 - April 2024

CEN3031 | University of Florida, Gainesville, Florida

- Utilized Agile frameworks to develop a web application, managing workloads into sprints and CI/CD development.
- Developed the backend of the application involving user authentication, database management, and OpenAI API calls

DGE (Diego Game Engine)

May 2021 - Present

- Developed good habits and learned best practices for improving code modularity in a big project.
- Learned many low level aspects of a game engine such as render pipelining by using only C++ and OpenCL.
- Integrated OOP build patterns, such as fluid builders, to streamline creation of many large scale backend systems within the project.

Itch.io Games

May 2020 - Present

- Strengthened familiarity with Unity and C# by designing and programming various games and participating in game jams.
- Planned and coded the implementations of various features in original games utilizing Unity Scripting API.

SKILLS

Programming Languages: C, C++, C#, Java, Javascript, Python, SQL, HTML, CSS, MATLAB, AutoHotkey, LaTeX, ARM Assembly, OpenCL, Makefile, CMake, Rust.

Other Skills/Awards: Command Prompt, Unix, Design Patterns, Machine Learning, Neural Networks, Unity, SDL2, SFML, Git, GitHub, GitHub Actions, Databases, Blender, Photoshop, Premiere, Jupyter Notebook, UF Dean's List.